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Experiment No: 01 Title:

String operations with and without pointers to arrays

Theory:

Strings are actually one-dimensional array of characters terminated by a **null** character '\0'. Thus a null-terminated string contains the characters that comprise the string followed by a **null**.

The following declaration and initialization create a string consisting of the word "Hello". To hold the null character at the end of the array, the size of the character array containing the string is one more than the number of characters in the word "Hello."

```
char string[6]={‘H’,’E’,’L’,’L’,’O’,’\0’};
```

C supports a wide range of library functions that manipulate null-terminated strings .Few are as follows:

1) **strcpy(s1, s2) :**Copies string s2 into string s1.
2) **strcat(s1, s2) :**Concatenates string s2 onto the end of string s1.
3) **strlen(s1) :**Returns the length of string s1.
4) **strcmp(s1, s2):**Returns 0 if s1 and s2 are the same; less than if s1<s2; greater than 0 if s1>s2

Pointers and strings: Strings can be stored at some memory location and assign the address of the string in a char pointer. This is shown as below:

```
char str[]=”Hello”; char
*p=”Hello”;
```

We can assign a char pointer to another char pointer.

```c
#include<stdio.h>
#include<string.h>

int main(){
   char s1[100],s2[100],s3[100];
   char *p;

   printf("Enter first string: ");
   scanf("%s",s1);

   printf("Enter second string: ");
   scanf("%s",s2);

   printf("\nLength of first string: %d",strlen(s1));

   strcpy(s3,s1);
   printf("\nCopied string: %s",s3);

   strcat(s1,s2);
   printf("\nConcatenated string: %s",s1);

   printf("\nComparison result: %d",strcmp(s3,s2));

   p=s3;
   printf("\nString using pointer: %s",p);

   return 0;
}
```

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Assignment 1

Q1 What is a string?

-> A string is a sequence of characters. In C/C++ it is represented either as a character array terminated by a null character('\0') or managed dynamically using the std::string class

Q2 Explaining string functn strcmp(), strlen(), strcat(), strcpy()

-> strlen(s): Calculate and returns the total no. of character in string s.

strcpy(dest, src): Concatenates the src string onto the end of the dest string.

Q3 In string comparisons, if two strings are same, what will be the result of comparision?

-> The strcmp() function returns 0 when both string are identical.

Q4 What is length of string data str?

-> the length of is -.

Q5 Explain how to use strcpy() function.

strcpy() is used by passing the destination char array as the first argument & the source string as the second argument. strcpy(destination, source)

Q6 What is strncat() string fn?

-> Strncat(dest, src, n) appends at most n characters, from the src string to the end of the dest string providing protection against buffer overflows compared to strcat().

Q7 What is strrev()?

-> strrev() is a non standard string handling function available in <string.h> on some platform compilers like MSVC that reverses the characters of string in place.

Q8 Applicath of (string)

- Text processing & document editing
- Database storage & queries handling
- Compiler design.